

NITIN KUMAR

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Summary

AI-focused Computer Science graduate driven by intelligent system design and real-world problem solving. Experienced in building machine learning and deep learning applications across NLP and Computer Vision domains. Engineered scalable ML-powered systems including spam detection pipelines, gesture-to-speech translation frameworks, and image caption generation models. Passionate about transforming complex data into deployable AI solutions using Python, modern ML frameworks, and optimized data pipelines. Committed to innovation, analytical precision, and building next-generation intelligent applications.

Education

Chandigarh University – Mohali, Punjab

Aug 2022 – May 2025

Bachelor of Engineering – Computer Science and Engineering

Projects

- **Intelligent Spam Classification System**

- Architected an end-to-end NLP classification pipeline using Python and Scikit-learn to detect spam in SMS and Email communications.
- Implemented TF-IDF (Term Frequency – Inverse Document Frequency) vectorization to transform unstructured text into numerical representations.
- Applied Naive Bayes algorithm for probabilistic classification with optimized preprocessing using NLTK (Natural Language Toolkit).
- Deployed the system using Streamlit to enable real-time prediction and interactive user testing.

- **Sign Language to Speech Translation System**

- Engineered a Computer Vision pipeline to translate real-time hand gestures into speech output.
- Integrated Deep Learning-based gesture recognition models for accurate classification.
- Designed real-time video frame processing workflows ensuring low-latency inference.
- Developed an accessibility-driven solution aimed at improving communication for speech and hearing-impaired individuals.

- **Real-Time Object Detection**

- Engineered a high-performance computer vision pipeline using Python and OpenCV to process live video feeds with low latency.

- Deployed the YOLOv3 architecture to identify and classify objects across 80+ categories with high confidence scores in real-time.
- Optimized inference speed using NumPy for efficient matrix operations, ensuring smooth frame-rate processing without lag.
- **Sentiment Analysis of Movie Reviews**
 - Engineered an NLP classification pipeline using Python and Scikit-learn to predict sentiment polarity in movie reviews.
 - Implemented TF-IDF vectorization to transform unstructured text data into numerical features for efficient model training.
 - Optimized data preprocessing workflows using Pandas for cleaning and tokenization, ensuring high accuracy for the Logistic Regression model.
- **Image Caption Generation using Deep Learning**
 - Developed an AI-powered captioning system combining CNN (Convolutional Neural Network) for feature extraction and RNN (Recurrent Neural Network) for sequence generation.
 - Integrated pre-trained deep learning models and fine-tuned them to enhance caption accuracy.
 - Optimized model training workflows and improved contextual sentence generation performance.
 - Demonstrated practical application of multimodal learning and sequence modeling techniques.
- **Image to Text Generator**
 - Developed an end-to-end OCR web application using Python and EasyOCR to extract and digitize text from unstructured image data with high accuracy.
 - Engineered a robust Flask API to manage image uploads and inference requests, integrating OpenCV preprocessing pipelines (thresholding, noise reduction) to optimize recognition performance.
 - Deployed the solution using Gunicorn WSGI server, ensuring scalable and reliable handling of concurrent user requests in a production-like environment.

Experience

Mobile App Development Intern - [Link](#)

Teachnook-Cognizance'24, IIT Roorkee

- Engineered a real-time News Application featuring category-wise content architecture and dynamic content rendering.
- Managed the complete software development lifecycle from requirement analysis to deployment.
- Implemented optimized UI/UX flows ensuring seamless navigation and responsive performance.
- Collaborated in cross-functional teams to deliver scalable and user-centric mobile solutions.

Skills

- **Generative AI & LLMs:** Generative AI, Large Language Models (LLMs), RAG Pipelines, Prompt Engineering, NLP, Transformers, Embeddings, OpenAI (GPT), Google Gemini, LangChain, Hugging Face
- **Programming & Backend:** Python, Java, C++
- **Machine Learning & AI:** NLP, Deep Learning, Computer Vision, Model Training & Evaluation

- **Professional Competencies:** Leadership, Strategic Thinking, Collaboration, Cross-Functional Collaboration, Problem Solving

Certifications

- Java as a Second Language – Coursera: [Link](#)
- AI for Everyone – Coursera: [Link](#)
- Introduction to Neural Networks and PyTorch – Coursera: [Link](#)
- The Java Language – Coursera: [Link](#)
- Mobile Development and JavaScript – Coursera: [Link](#)

Honors

- Successfully led and coordinated team-based technical projects, ensuring timely delivery and efficient task management.
- Awarded certifications in Data Visualization and Deep Learning.
- Successfully led and coordinated team-based technical projects.